The Wrong Unit of Uncertainty: Simultaneous Inference for Repeated Cross-National Surveys

Kunwoo Park

Department of Politics, Kookmin University, Seoul, Republic of Korea

pkw31386094@gmail.com ORCID: 0009-0007-9067-8964

Abstract

Uncertainty should attach to the unit of the claim; repeated cross-national surveys attach it to the wrong unit twice: a wave-pair contrast is read as a persistent, distribution-wide trend, and the estimated distributions it calibrates on are treated as truth. For the claim unit, a deployable instrument: a finite-sample simultaneous band over a country's whole response-distribution *trajectory*, the country the exchangeable unit, a graded claim family on one band—valid at any K for the estimated trajectory, bounded for the latent one, inside a selector whose nested fallbacks preserve validity under any rule. For the calibration unit, a boundary: the latent correction is non-identified without the design-noise law and unreachable across countries with it, yet it activates at the small-area unit of the same survey, cutting the fallback band by a certified 16–22%. In the ESS it cuts a marginal reading of twenty of thirty countries to net decline in six. Over 2002–2024 none certifies a persistent slide, a rung only crisis-scale change clears; twenty-three of thirty-three certify both falls and rises, and span erosion certifies in eight. On the World Values Survey the same rung shift cuts the certified set 1.9–4.8×; the survivors are post-communist and Arab-Spring states.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament in a country and year is a cumulative distribution function over a response scale. Repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey, the AmericasBarometer (Cohen et al., 2021), let us track how that distribution moves across waves and *transport* it: predict one country's trajectory from the others. Conformal methods are a natural tool, promising finite-sample, distribution-free coverage.

Two wrong units. Uncertainty should be attached to the same unit as the substantive claim; claims from repeated surveys violate this twice. The *claim* "democratic trust is declining" is typically built from a single wave-pair mean contrast, then narrated as a country-level trend — our instrument instead covers the whole trajectory at once, with a partially ordered family of claims read off one band, whose top rung is strictly stronger than the net trend the literature typically intends (Section 7). The second, subtler wrong unit is the calibration: the objects conformal transport calibrates on are themselves *estimated* from complex probability samples, and standard practice plugs them in as truth.

The two errors are distinct. Writing $F_{c,r}(t)$ for country c's latent CDF at threshold t in wave r and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A band calibrated on the observed $\widetilde{F}$ scores treats the sum as the transport error alone: when the design error is non-negligible the band is too wide, yet any correction subtracting the survey noise can make it too narrow, being itself estimated from few countries.

Contributions. We make four. *First, the claim-unit problem, and an instrument for it.* We separate the claim unit (a wave-pair contrast standing in for a distribution-wide trajectory) from the calibration unit (the omitted design-error layer of (1)), and give a finite-sample simultaneous band over a country's whole trajectory whose partially ordered claim family is read off one coverage event. The two layers get two instruments: within-country *certification* is a design-based simultaneous sup-t band over the wave-pair contrast surface, while conformal transport calibrates cross-country prediction — the theory's finite-sample language attaches to the transport band, not to the certification counts. *Second, an impossibility theorem for estimated calibration objects.* Stated at the curve level of (1): without information on the design-noise law the latent transport-error law is non-identified, and no band built on the observed curves can uniformly improve on the plug-in conformal band while remaining valid (Theorem 1). Estimated calibration is not a nuisance to correct away; it is an inferential boundary. *Third, the feasibility boundary of the correction, and a procedure that respects it.* Even with the design-noise information supplied, the reliability diagnostic obeys an algorithmic floor $D \geq \sqrt{2/(K-1)}$, so the deployed correction requires $K \geq 94$ exchangeable populations; the two largest surveys fail in opposite ways — the WVS clears the size barrier ($K \approx 100$) with negligible design noise, the ESS has genuine clustered noise but $K \leq 33$. The deployed selector is the theory's consequence: its anchor band is exact at any K under its symmetric centers (conservative to $O(1/K^2)$ as deployed, Theorem 3), its anchor branches are nested so selection among them is validity-free, and the non-nested deconvolution branch is charged a frozen budget only where its gate can open (Theorem 5). One unit down, on a region's departure from its national distribution ($K \approx$ 230–290), the same frozen selector does activate, and holding out whole countries confirms the region is the exchangeable unit (Section 6). (The clustered band is machinery shared with a companion paper on a non-survey domain, Park, 2026; new here is what the survey layer forces — the impossibility, the floor, the safe selector, and the claim-family deployment.) *Fourth, two named reanalyses where the instrument changes the substantive picture* (Section 7). For

European trust the family certifies net decline in six of thirty (2018–2024), not the twenty flagged marginally. Over the full 2002–2024 record no country certifies persistence, twenty-three of thirty-three certify both declines and recoveries, and eight certify span erosion, all read off one band at one level with no multiplicity or endpoint-selection corrections. Globally, applying a trajectory-persistence criterion to the Foa–Mounk "deconsolidation" battery on the WVS/EVS (1981–2022) cuts the wave-pair certified set by 2.6–6.5×. Aggregating per-item certifications then turns the correction into a discovery: the thirteen-country *certified core* concentrates in post-communist and Arab-Spring-aftermath states.

Sections 2–9 proceed in that order: setup and impossibility, method, theory, validation, the real-data regime, the reanalyses, related work, scope.

2 Setup and the two targets

The object we deploy is the exact simultaneous band of Theorem 3; the impossibility below (Theorem 1) is a scope boundary delimiting when a design-aware correction to that band is even available.

Objects. We observe K source countries, indexed $c = 1, \ldots, K$, treated as exchangeable units (the "population" or cluster is the exchangeable object, not the individual respondent). Each country has a latent *response-distribution trajectory* $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \ldots, t_T\}$ and survey rounds r. We say *response distribution* deliberately: guarantees attach to recorded answers, and reading a certified shift as an *attitude* shift additionally requires measurement invariance across waves (Section 9). One invariance comes free: every claim below depends on responses only through threshold events, so nothing changes under monotone recoding of the ordinal scale. We do not see $F_{c,r}$; we see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified-PSU / replicate-weight bootstrap can estimate. Define the transport score of a country as its worst-case standardized deviation from the transport

Procedure	Target	Randomness	Assumptions	Guarantee
Clustered PCB (anchor)	T1 $\widetilde{F}_{\text{new}}$	country exch.	—	exact $\geq 1-\alpha$,
same band, latent reading	T2 F_{new}	+ design noise	noise tail, anti-conc.	$1-\alpha-$gap, ga
Conservative envelope	T1	country exch.	nested $\supseteq$ PCB	inherits the a
Deconvolution branch	T2	+ design bootstrap	(A1)–(A3), $K \geq 94$	$1-\alpha-\varepsilon_{K,B}$; fr
Certification sup-t (§7)	contrast surface $D_{a,b}(t)$	survey design	bootstrap regularity	asymptotic si

Table 1: Which procedure guarantees what. T1 is the survey-estimate trajectory, T2 the latent one; the certification instrument of Section 7 is design-based and within-country, and the theory's finite-sample language does not transfer to it.

center,

$$R_c \;=\; \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; R_c + \xi_c,$$

where $\widetilde{R}_c$ is the same quantity computed from the observed $\widetilde{F}$ and ξ_c is the induced design contamination. (Because R is a supremum the additive centred score model is a working approximation; the impossibility below is stated at the curve scale and does not use it.) The governing scalar is the design-to-total SD ratio

$$\rho \;=\; \frac{\mathrm{SD}(\xi)}{\mathrm{SD}(\widetilde{R})} \;=\; \sqrt{\frac{\overline{v^2}}{s_R^2 + \overline{v^2}}} \;\in\; [0,1),$$

$\rho \to 0$ is the design-noise-negligible regime, $\rho \to 1$ the noise-dominated one. (The simulations' dial is the unbounded $\mathrm{SD}(\xi)/\mathrm{SD}(R)$; reported values use the bounded definition.)

Target and validity. We want a simultaneous band covering a new country's *latent* trajectory, $\Pr\big(F_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\big) \geq 1-\alpha$. Ordinary clustered conformal on the observed scores is exact for the *survey-estimate* target $\widetilde{F}_{\text{new}}$, needing only exchangeability; its latent coverage differs by an explicit gap vanishing with the design-noise scale (Theorem 2(b); symmetry alone is not enough), its width is inflated by design noise, and correcting that is a deconvolution. Table 1 collects every procedure–target–guarantee triple in the paper; each later section fills in one row.

What country exchangeability assumes. The transport band's identifying assumption is a superpopulation postulate: participating countries' trajectory laws are symmetric draws, over a set that is not sampled and enters rounds non-randomly — a strong idealization, tempered three ways. The headline certifications of Section 7 do not use it, being within-country design-bootstrap statements; under approximate exchangeability the Barber et al. (2023) bound controls the gap by an unobserved total-variation distance, treated as qualitative; and a leave-one-region-out audit finds near-nominal held-out coverage except for the post-communist East (6/9 and 5/9 at nominal 0.90)—a quantified scope condition on transport *into* that bloc (Supplementary Material).

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated curves alone. The result is stated at the curve level of (1), where the additive model *is* the survey sampling structure; nothing is assumed about the sup-scores, and the design noise may be arbitrarily heteroskedastic across thresholds.

Theorem 1 (Non-identification and necessity of design information)**.** *Let the observed deviation curves be* $\widetilde{E}_c = E_c + S_c$ *as in* (1)*, the latent curves exchangeable with law* $\mathcal{L}_E$*, the design-noise curves independent of them with law* $\mathcal{S}$ *in an admissible class that contains* $S \equiv 0$ *and is closed under one further nondegenerate symmetric convolution. (i)* $\mathcal{L}_E$ *is not identified from the observed law* $\mathcal{L}_E * \mathcal{S}$*. (ii) Any band whose radius is measurable in the observed curves alone and is valid for the latent target over the whole admissible class cannot uniformly improve on the plug-in conformal band: on the member* $S \equiv 0$*, which it cannot distinguish from the contaminated configurations, the plug-in radius is already the tightest valid one up to the conformal discreteness slack, which randomized smoothing exhausts. Uniform improvement requires shrinking the class of noise laws.*

The reading is constructive (proof in Appendix A): a band that is both valid and narrower than the plug-in must supply the design-noise law—replicate weights or a design bootstrap, exactly what Section 3 uses.

Remark 1 (Beyond surveys). Nothing here is survey-specific: the result holds whenever conformal calibration uses estimated objects: MRP small-area outputs, learned or imputed features, meta-analytic effects. Design/measurement information is the identifying ingredient, and its reliability at finite K is itself bounded (Proposition 1); confirmed outside surveys on a small-area calibration (Supplementary Material).

3 A nominal-safe adaptive method

Theorem 1 says the design bootstrap is the identifying input. The method turns that input into a band by choosing, target-blind, among three constructions and by never using the aggressive one unless it is provably reliable at the available number of countries.

Three branches. For calibration error curves $\{\widetilde{E}_c\}_{c=1}^{K}$ over threshold grid t, with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap: *(i) Clustered PCB* takes the conformal quantile of the *unstudentized* scores $\max_t |\widetilde{E}_c(t)|$ (the U0 construction): exact for the survey-estimate target at any K, with the latent gap of Theorem 2(b). *(ii) Deconvolution* studentizes by the finite-K-safe target scale $\widehat{s}_T^2(t) = \max\big(s^2_{\text{plug}}(t) - [\widehat{\overline{v^2}}(t) - z\ \text{SE}]_+,\ \text{floor}\big)$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1-\rho^2}$ relative to PCB. *(iii) Conservative* inflates each unstudentized score by the design SD, $\max_t(|\widetilde{E}_c(t)| + z\,\widehat{v}_c(t))$, a score that dominates the PCB score pointwise, so the conservative band *contains* the PCB band by construction. This nesting is load-bearing: selection between nested bands anchored at an exact band is validity-free, so the selector needs no conditional-coverage argument.

Safe selector (Algorithm 1). A naive selector asks *can we deconvolve* and *do we need to*; the safe selector adds the missing third question, *can we do it safely at this K?* Let $\widehat{\rho}_{\text{LCB}}$ be a conservative lower bound on ρ, $D = \max_t \text{SE}(\widehat{s}_T^2)/\widehat{s}_T^2$ a reliability diagnostic, and $\widehat{\delta}_{\text{UCB}}(D)$ a frozen upper bound on the finite-K coverage remainder. Deconvolution is used

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{R}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, target curve $\widehat{F}_{\text{new}}$, level α; frozen $\rho_0, \delta_{\max}, g_{\min}, \Delta_g$ and the K-only budget rule $\alpha_{\text{dec}}(K)$

set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);

compute diagnostics s, $\widehat{s}_T$, $\widehat{\rho}_{\text{LCB}}, D, \widehat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{PCB}}, r_{\text{con}}$ at level α_{anchor}; and r_{dec} at level α_{dec};

if $\widehat{\rho}_{LCB} \leq \rho_0$ **then**

  branch $\leftarrow$ PCB;

else if $\widehat{\delta}_{UCB}(D) \leq \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**

  branch $\leftarrow$ deconvolution;

else

  branch $\leftarrow$ conservative;

return $\widehat{F}_{\text{new}} \pm r_{\text{branch}}$, branch, and diagnostics;

iff all four gates pass: (A) $\widehat{\rho}_{\text{LCB}} > \rho_0$ (design noise materially large); (B) $\widehat{\delta}_{\text{UCB}}(D) \leq \delta_{\max}$ (remainder within budget); (C) $\widehat{\text{Gain}} - \Delta_g > g_{\min}$ (beats conservative by a margin); (D) stability. Otherwise the band is PCB when $\widehat{\rho}_{\text{LCB}} \leq \rho_0$ and the conservative envelope when design noise is real but information insufficient. Every gate is a function of the calibration set only, so the selector is target-blind, and by Theorem 5's nested-band argument, validity survives even adversarial selection between the anchor branches. The deconvolution branch, which is not nested, is charged its own miss budget, and only in the $K \geq 94$ regime where its gate can open at all (Section 4 states the rule). All constants are frozen before validation (Appendix B); when deconvolution is used the reported level is the observable $1-\alpha-\widehat{\delta}_{\text{UCB}}(D)$, a frozen *calibrated* bound (the theorem-level guarantee is Theorem 4's $\varepsilon_{K,B}$).

Deployment. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, returning the band, the selected branch and why, the coverage level, and the target that level refers to.

4 Theory

We deploy Theorem 3 (the base band, exact at every K) and Theorem 5 (the same guarantee through the selector, under any selection rule); Theorems 2 and 4 price what the deployed band does not get for free — the latent rather than estimated trajectory, an estimated rather than known noise law. Proofs are in Appendix A; each theorem is paired with a machine-checked contract test.

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is known. (a) The clustered band is finite-sample valid for the survey-estimate target,* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq 1 - \alpha$*, using exchangeability only. (b) For the latent target the same band satisfies, for every* $u > 0$, $\Pr(F_{new} \in \widehat{B}) \geq 1 - \alpha - \Pr(\xi < -u) - \Pr(\widetilde{R} \in (\widehat{q} - u, \widehat{q}])$*. (c) The deconvolution band with the true target scale is valid for the latent target, and finite-sample exact under the scale-family assumption (A1), with width a factor* $\sqrt{1-\rho^2} + o(1)$ *of the clustered band.*

The gap in (b) is a noise-tail plus a local anti-concentration term, vanishing with the design-noise scale at rate $\rho^{2/3}$ under variance-only tails and ρ up to logarithms under sub-Gaussian noise — but not by symmetry (a bimodal counterexample in the Supplementary Material), which is why we estimate and report ρ rather than assume it away.

Theorem 3 (Exact finite-K validity of the deployed base band)**.** *Let the base band use the unstudentized cluster score* $R_c = \max_t |E_c(t)|$*, with a center satisfying the symmetric-construction condition (Appendix A states the four admissible cases). If the countries are exchangeable, then for every K the band* $\widehat{B} = \mathit{center} \pm \widehat{q}$*, with* $\widehat{q}$ *the* $\lceil (1-\alpha)(K+1) \rceil$*-th order statistic of* $\{R_c\}$*, satisfies* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq \lceil (1-\alpha)(K+1) \rceil / (K+1) \geq 1 - \alpha$*, distribution-free.*

The guarantee is exact for the survey-estimate trajectory T1; the latent trajectory T2 adds the gap of Theorem 2(b), bounded through the reported ρ. Exactness at every K is bought by excluding two asymmetries a more adaptive band would introduce, in-sample modulation and grand-mean centers containing their own unit, at a cost of at most 5% of width on the near-homoskedastic real curves.

Theorem 4 (Estimated-law validity)**.** *Assume a common scale family indexed by the design variance (A1), anti-concentration of the sup-score law near its upper quantile (A2), and bounded curves with an unbiased size-B design bootstrap (A3); the appendix states all three precisely. Then (i) with oracle scales the deconvolution band is finite-sample exact for the latent target at every K, and (ii) with estimated scales it satisfies* $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha-\varepsilon_{K,B}$ *with* $\varepsilon_{K,B} = O\big(\sqrt{\log(KT)/\min(K,B)}\,\big)$.

The remainder's direction is fixed by construction: the finite-K-safe scale $\widehat{s}_T^2$ of Section 3 subtracts a lower confidence bound on the design variance, so outside an α_2-probability event the scale error widens the band. Some shape restriction is not removable (the counterexample above); (A1) is one sufficient choice.

The deployed pipeline. The miss budget is a frozen function of K alone: below the reliability floor ($K < 94$, the regime in which the gate cannot open at all, Proposition 1, and the one covering every certification sample here including the WVS $\geq$ 2-wave sets at $K = 59$–77) the anchors keep the full α and the deconvolution budget is zero; at $K \geq 94$ the budget $\alpha_{\text{dec}} = \min\{\max(0.1\alpha, 3/(K{+}1)), \alpha/2\}$ is reserved for it (derivation in the Supplementary Material).

Theorem 5 (Safe-adaptive finite-K validity)**.** *Deploy Algorithm 1 with the unstudentized PCB and conservative branches, scores* $\max_t |E_c(t)|$ *and* $\max_t(|E_c(t)| + z\widehat{v}_c(t))$*, at level* α_{anchor}*, and the deconvolution branch at* α_{dec}*. Then, finite-sample and with no condition on the selection rule,*

$$\Pr\left(\widetilde{F}_{new,r}(t) \in \widehat{B}\right) \geq 1-\alpha \quad (K < 94), \qquad \Pr\left(F_{new,r}(t) \in \widehat{B}\right) \geq 1-\alpha-\varepsilon_{K,B} \quad (K \geq 94).$$

The clauses name different targets because they rest on different assumptions. Exchangeability alone delivers T1 exactly, with T2 adding Theorem 2(b)'s gap; (A1)–(A3) deliver the latent target directly, by an anchor-domination lemma, and the remainder $\varepsilon_{K,B}$ they carry is charged to the deconvolution branch alone.

The proof uses no conditional-on-selection step. A nested-band lemma does the work: the conservative score dominates the PCB score pointwise, and selection between nested bands anchored at an exact band is validity-free. That is nested-conformal machinery (Gupta et al., 2022; Yang and Kuchibhotla, 2024); our contribution is to engineer the branches so that it applies. A union bound then charges the one non-nested branch its own budget. Two scope notes. At $K \geq 94$ the reported level is a frozen calibrated bound, not a theorem. And the unsurveyed-target deployment uses a leave-one-out center, outside Theorem 3's symmetric constructions: its residual asymmetry is $O(1/K^2)$, dispersion-ordered in the conservative direction (a heuristic for sup-scores, not a bound; Supplementary Material), and a 28-cell simulation across four families finds no cell below the exact-construction floor beyond Monte Carlo error (`e58`). The strictly exact alternative is infeasible at survey scale: halving the calibration set below $\lceil 1/\alpha \rceil$ units makes the conformal radius infinite, exactly at the K this paper works at.

Efficiency. The low-ρ deconvolution/PCB width ratio is $1 - \frac{1}{2}\rho^2 + O(\rho^4)$, proved; conservative $\supseteq$ PCB holds pointwise; and selection cannot inflate miscoverage (Theorem 5). The remaining width results, one of them open, are in Appendix A.

5 Simulation and a design-preserving stress test

Simulation establishes two facts: attaching the band to the wrong unit collapses coverage, and the safe selector meets its guarantees on data it has never seen.

5.1 The wrong unit collapses coverage

On a ground-truth simulation ($K = 30$ countries, $T = 6$ thresholds, errors correlated across rounds and thresholds) we ask each band for *trajectory* coverage, the whole curve-path at

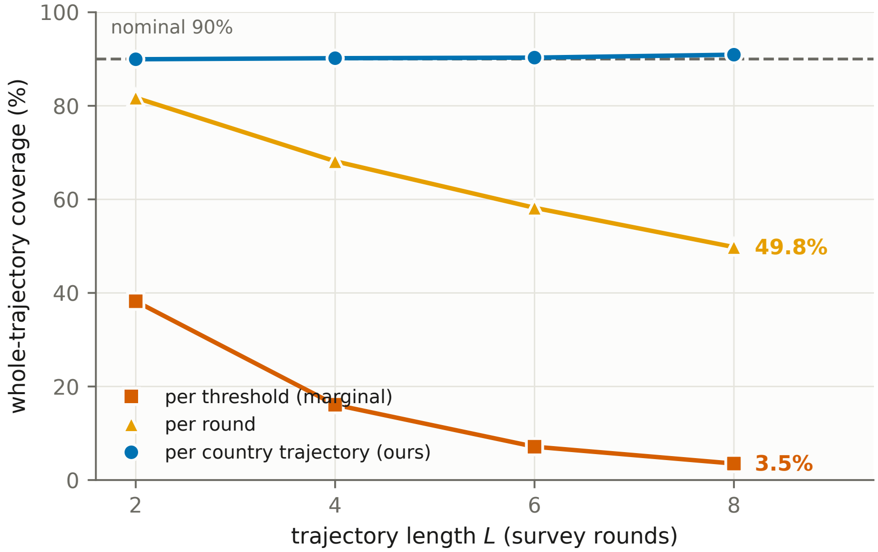

Figure 1: Coverage of a whole country trajectory, by the unit the score is attached to (4,000 replicates, nominal 90%). The gap widens with the length of the trajectory being claimed about.

once (Figure 1). All three bands are unstudentized and finite-sample, so only the score's unit varies. A band that is marginally valid at the round level covers the trajectory 81.7% of the time at $L = 2$ and 49.8% at $L = 8$; per threshold the figures are 38.2% and 3.5%. Only the country-trajectory band holds its nominal 90% at every length, because the trajectory is a single exchangeable unit. This is the coverage face of the paper's thesis, and the same mechanism drives the certification-count collapse of Section 7. Pre-registered, reported as produced (`e28`).

Severity: what each rung can detect. At ESS-like design noise (`e32`), 80% power at the *net* rung needs a per-pair decline of 0.02–0.03 CDF points (realistic secular erosion); the *persistent* rung needs 0.06–0.08, crisis-scale, and its power *falls* as the trajectory lengthens. Injecting known declines into the real ESS pairs, each estimate drawn from that country's own stratified-PSU bootstrap, puts the thresholds at ≈ 0.033 (net) and ≈ 0.075 (persistent), with realized size 0.001–0.007 against a nominal 0.10: the rule is markedly conservative, and non-certification is weak evidence of absence at every rung (`e42`). The powered instrument for secular decline is the net rung, co-headlined in Section 7.

Validation, in three disclosed layers. The frozen procedure passed three validation layers, each disclosed in full in the Supplementary Material: a first sealed holdout demoted to *development* after an audit found its scorer evaluating a band other than the deployed one; a corrected-scorer rerun of the same grid (450/450 cells floor-compatible, worst 0.882); and, because both layers had touched the final design, one further sealed run on six unseen DGP families, script hash recorded before first execution and reported as produced including a disclosed generator bug: 120/120 floor-compatible, worst 0.878, zero activations below $K = 94$ (`e33`).

A design-preserving stress test, and what it does not show. Subsampling the real AmericasBarometer, the deployed selector never fires, and the pooled $\widehat{\rho}$ turnover at deep subsampling coincides with $\approx$ 1 PSU per stratum, where the m-of-m variance estimate collapses — a failure mode of the $\widehat{\rho}$ diagnostic under sparse-PSU designs, not a saturation law; the regime characterization of Section 6 rests on full-sample estimates (Supplementary Material).

6 Evidence from the ESS and the AmericasBarometer

The regime characterization. On real data the selector reduces to clustered conformal, and the reason is worth stating. Across both surveys, both estimands, and three outcomes, cross-national inference is a uniformly *low design-noise* regime: $\widehat{\rho} \leq 0.22$ on LAPOP, at most 0.29 in the ESS subgroup scan, well below the $\rho_0 = 0.47$ cutoff. Every $\widehat{\rho}$ we report is a downward-biased *floor*, because the PSU bootstrap resamples m-of-m without Rao–Wu–Yue rescaling (Rao et al., 1992) and single-PSU strata contribute no variance. The rescaled rerun confirms the floors are tight: maxima 0.281 (ESS) and 0.222 (LAPOP), still severalfold below ρ_0, zero gate activations, every certification count unchanged; and singleton strata, which no rescaling repairs, are 1 of 7,028 in the core rounds. The safe selector thus reduces to clustered PCB on all real data, not by assumption but by its own low-ρ gate.

Why the deconvolution is unreachable, precisely.

Proposition 1 (Survey-scale unreachability)**.** *The deployed selector activates deconvolution only if a* need *gate* $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ *and a* reliability *gate* $D \le \tau_D$ *both hold,* $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$*. Both are bounded by* algorithmic identities of the frozen procedure. *(a)* $\rho = \sqrt{\overline{v^2}}/s_{\mathrm{plug}} \in [0, 1)$ *with* $s_{\mathrm{plug}}^2 = s_R^2 + \overline{v^2}$*: the need gate fires only when design noise is an appreciable fraction of the between-population signal. (b) Since* $\widehat{s}_T \le s_{\mathrm{plug}}$ *by construction,* $D \ge \sqrt{2/(K-1)}$*, so the reliability gate requires* $K \ge 1+2/\tau_D^2$*, i.e.* $K \ge 94$ *at the frozen* $\tau_D = 0.147$*. (Appendix A separates the deterministic statement from its kurtosis-dependent interpretation.)*

The two requirements pull apart on real surveys (Figure 2). On the ESS the noisiest subpopulations raise $\widehat{\rho}$ only to 0.29 while $K \le 33$ leaves $D \ge 0.25$: neither gate opens (42 cells, zero activations). The WVS is the mirror image: its full item-coverage sets ($K = 95$–105) make the reliability gate feasible, the certification samples ($K = 59$–77) do not, and in either denominator the weights-only $\widehat{\rho}_{\mathrm{LCB}} \le 0.09$ fails the need gate decisively. The survey large enough to estimate the deconvolution has almost no design noise to remove; the survey with real clustered noise has far too few countries.

Where the correction does live: small areas, on this same survey. The barrier is about the unit, and the ESS itself contains a finer one. Take the estimand this regime is actually for, a region's departure from its own *national* distribution, $D_g = F_g - F_{\mathrm{nation}}$, and calibrate the transport band across regions (`e54`). A single stratified-PSU bootstrap of the whole country recomputes F_g and F_{nation} on every replicate, so the design SD is that of the difference (exclusion rules in the Supplementary Material).

On that estimand the design-to-total ratio is not marginal: $\widehat{\rho}_{\mathrm{LCB}}$ runs 0.26–0.52 and clears $\rho_0 = 0.47$ in eight of twenty-three unit-rounds. Reliability needs scale, and the two gates open together only once $K \gtrsim 230$: units of forty to sixty respondents, rounds 9 and 10, $K = 228$–287. There the *frozen* selector, no constant retuned, activates the design-aware branch for the first time on real survey data: a band 20.5–26.6% narrower than the conservative envelope at the reported level 0.881, with certified width-gain lower bounds of 0.16–0.22 against the 0.10 efficiency threshold, across twelve independent bootstrap streams; the deconvolved target scale itself sits 15–17% below its plug-in value.

Is the region an exchangeable unit? The regions nest in twenty-seven countries, nine each, and between-country differences carry 32–49% of the variance in a region's sup-departure: the objection this paper would itself raise is that the effective unit is the country and K is really 27. Holding out whole countries answers it (`e55`). Calibrating on the other twenty-six countries' regions, the plug-in anchor covers 0.893–0.903 against its 0.900 target in all twelve cells, and, the informative contrast, differs by at most 0.015 from leave-one-*region*-out calibration, which keeps the held-out region's country-mates. Deleting nine country-mates from the calibration set costs nothing, so within-country dependence does not deflate the quantile. In the activating cells the selected band covers 0.94–0.98 against its nominal 0.88, scored on the observed departure, hence a lower bound for a branch targeting the latent one. Coverage conditional on the country does not transfer: the median held-out country is fully covered but the tenth percentile is 0.75, the worst three (Bulgaria, Greece, Hungary, 0.53–0.63) being those with the largest regional heterogeneity. The guarantee is marginal over regions, as conformal guarantees are, and we do not read it per country.

Two further scope conditions travel with it. Reaching $K \gtrsim 230$ requires pooling regions coded at different NUTS levels; restricted to the fourteen countries sharing one level the need gate still opens in nine of twenty-one unit-rounds, but $K \leq 140$ keeps the reliability gate shut and the selector falls back to the conservative envelope, the same two-barrier structure as the national result above, one level down. And round 11 never activates, because its regional design noise is lower. The correction is therefore not unreachable in principle, and

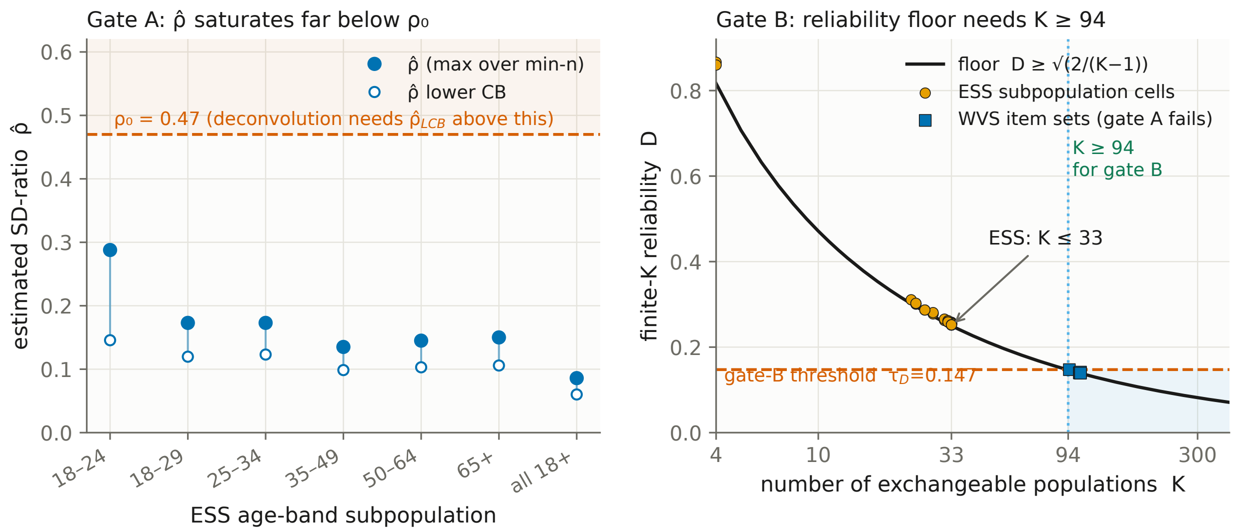

Figure 2: The two barriers at the national unit, and why no survey clears both. Left, $\widehat{\rho}$ against ρ_0 by ESS age band (each point the maximum over that band's min-n sweep, 42 cells). Right, reliability D against its floor $\sqrt{2/(K-1)}$: the ESS sits on the floor at $K \leq 33$, and only the WVS item sets reach $K \geq 94$, where gate A fails instead.

not reachable across countries: it lives at the small-area scale, which is where Theorem 1's remark said it would.

How coarse can the calibration unit be? One constraint is sharp and, we believe, unremarked: the conformal level is infeasible whenever $K < \alpha^{-1} - 1$, because the required order statistic exceeds the sample and the band is $[0, 1]$ at any width—a too-coarse unit is unusable, not merely inefficient, and the cross-national default clears the floor comfortably ($K = 30$–34 against nine). Refining is not free either: with units of a controlled target size built by grouping whole PSUs, the band's half-width falls monotonically as units coarsen, from 0.190 at forty respondents per unit to 0.065 at three hundred and fifty (`e52`). Both are statements about the *calibration* unit; an estimand's unit is fixed by the question, and choosing among substantive areal partitions is the modifiable areal unit problem (Openshaw, 1984; Robinson, 1950), to which we add nothing.

7 Political reanalysis: how many democracies really declined?

We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (2018–2024, the window whose integrated files ship PSU/stratum identifiers), then extend every trajectory to the full record (rounds 1–11, 2002–2024), asking at each rung how many countries survive. The comparative literature has grown skeptical of a continent-wide trust decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022); we add a formal instrument that says so and the wrong-unit diagnosis of why the over-reading arises — the dominant trend estimates (Valgarðsson et al., 2025) rest on latent-trait models with no design-based uncertainty layer (Tai et al., 2024), and their data end where our window begins.

A family of claims, and one band beneath it. The claims differ in their unit. A *pairwise* claim asks whether a given adjacent pair of waves shows a distributional shift; *any-pair* asks whether some adjacent pair does; *net* asks whether the first-to-last span declined; *persistent country-wide* asks whether the whole distribution declined simultaneously across the trajectory; *Bonferroni* further controls the number of countries examined. The rungs form a partial order, not a chain. Persistent implies both net (adjacent declines telescope to the endpoint span) and any-pair; but net and any-pair are incomparable — a dip that recovers certifies an adjacent pair with no net decline, and a slow accumulation can certify a long span while no single adjacent pair clears the critical value; both patterns occur in the data below. The remaining steps differ in kind: from a marginal pairwise reading to any-pair the *claim* gets no stronger, the *guarantee* does; Bonferroni strengthens along a different axis, the number of countries. No chain is needed, because the rungs are not separate tests: each, every sub-span, and the reverse direction is a function of one object, the country's surface of wave-pair contrasts, so all are certified on one coverage event. Reading one rung as another is the over-detection this paper is about.

Proposition 2 (Claim-family transfer)**.** *Let $\widehat{B}$ be a simultaneous $1-\alpha$ band for a country's contrast surface $\theta = \{\theta_{a,b}(t)\}$ over every ordered pair of observed rounds, and let $\{A_j\}$ be any family of claims, each a set of surfaces. Define the certified subfamily $J = \{j : \widehat{B} \subseteq A_j\}$. Then $\Pr\big(\theta \in A_j$ for every $j \in J\big) \geq 1-\alpha$, with no dependence on $|J|$: on $\{\theta \in \widehat{B}\}$, set inclusion delivers every certified claim at once.*

The proof is the line just given, and the argument is standard: reading each deterministic consequence off one simultaneous region is the post-hoc logic of Scheffé (1953), closed testing (Marcus et al., 1976), and the selection-free bounds of Goeman and Solari (2011); Katsevich and Ramdas (2020). The contribution is its deployment: trend claims in this literature are made rung by rung, and reading them off a single studentized sup-t band over the whole contrast family removes the corrections a per-rung construction needs: no Bonferroni across a country's pairs, no separate budget for certified recoveries, and no endpoint-selection cost. The price is a wider critical value, and we report what it costs.

Which instrument certifies what. Within-country *certification* is a design-based simultaneous (sup-t) band on wave-pair CDF differences from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007), an asymptotic guarantee in which exchangeability plays no role; the conformal transport band calibrates cross-country prediction and the regime diagnostics. Headline counts are certification counts, and the theory sections' finite-sample language does not transfer to them.

The counts collapse up the hierarchy. For `trstprl` over rounds 9–11 (Figure 3), a marginal pairwise reading flags 20 of 30 countries; an any-pair design-aware band drops this to 12; net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands); and the *persistent, design-aware simultaneous* band over the preregistered low-trust core (the strongest rung) certifies one, Greece, which also survives

Bonferroni across countries. One disclosure is essential. *Greece fielded only rounds 10 and 11 here*, so its trajectory is a single pair and the top rungs coincide for it (as for Czechia and Israel, which do not certify). Among the 27 two-pair countries none certifies persistence, while the United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent rung and are demoted by the design-aware layer alone. Both numbers are the headline: net decline in six of thirty, persistent decline in one, on a single pair. The severity analysis (Section 5) explains why: the persistent rung has 80% power only against crisis-scale declines, so its non-certifications are weak evidence of absence. For `stfdem` the pattern repeats (23 marginal, 14 any-pair, 8 net, one persistent: Greece, again one pair, failing Bonferroni there). The literature reads Greece as a crisis-era legitimacy collapse (Armingeon and Guthmann, 2014; Torcal, 2014), but those accounts concern 2009–2015 while the certified pair is 2020/21→2023/24.

The long window: the secular question, asked properly. Extending every trajectory to its full record (rounds 1–11; median six pairs, up to ten; rounds 1–8 weights-only) reframes the twenty-year narrative in both directions (`e36`). Zero of the thirty-three countries with three or more usable rounds certify persistent distribution-wide decline over their full record, plug-in included. The null is not a bare one: reversing the inequality certifies *recoveries*, and across the 1,173 ordered contrasts of the trust series certified movement is directionally balanced (193 declining against 212 rising). Inside the same band, and therefore at the same α for that country, twenty-three of thirty-three countries certify at least one decline and at least one recovery, on each outcome separately and twenty on both (`e50`); the two directions share one critical value by construction, so no within-country correction is needed. The count across countries is a conjunction of thirty-three coverage events and carries no familywise control, as do all country counts here unless the Bonferroni rung is named. Whatever European trust did over two decades, it was not a monotone slide, though we stop short of calling it episodic, which would additionally require same-signed runs and returns toward baseline.

Under Proposition 2—one band, every ordered contrast and both directions at $\alpha = 0.10$ jointly (`e50`)—net erosion over the whole span certifies in eight countries (Cyprus, Spain, the United Kingdom, Greece, Hungary, Israel, Italy, Ukraine; Table 2). Against the per-rung construction the joint statement costs France at the span rung ($9 \to 8$) and six countries at the any-pair rung ($28 \to 22$); the two runs define adjacency differently, and the persistent zero holds under both (Supplementary Material). The under-detection face of the wrong unit shows in the ledger of contrasts: Hungary certifies its 22-year span while only 9 of its 55 ordered contrasts certify, and the United Kingdom 11 of 55, erosion that a reading built from adjacent waves alone would mostly miss. Because every endpoint choice is certified at once, the band also delivers the *share of a country's ordered contrasts that certify a decline*: a lower bound, comparable within a country but not across (the shared critical value grows with record length, correlation 0.96; the denominator grows quadratically), so we do not rank on it. Within a country it is informative: Spain certifies over 32 of its 55 contrasts and Slovenia over 21 of 55, neither entering the eight because their span contrast happens not to certify, which is the endpoint sensitivity the joint band exposes. Ukraine's certified span decline of at least 25 points runs from the post-Orange-Revolution wave of 2005 to a wartime sample: a certified statement about those two waves.

The reclassified countries are the weak-signal, higher-noise cases—*certified by the marginal plug-in but inconclusive under the design-aware band*, never "false positives, " a term we reserve for simulation with known truth.

Two confounds this window cannot exclude. Round-10 fieldwork coincided with the pandemic's rally-round-the-flag movements, so "persistent decline through this window" is a claim made against a rally; and nine countries switched to self-completion at round 10, confounding wave-pair shifts with mode effects, a bias no design bootstrap absorbs. A mode audit built from the data's own mode variable bounds it: the Greek pair is mode-constant

Table 2: Certified net decline in parliamentary trust, 2002–2024, read off one band covering every ordered contrast and both directions at $\alpha = 0.10$ jointly. *Decline* $\geq$: simultaneous lower bound over the preregistered core, in CDF points. *Contrasts*: ordered round pairs certified declining, of all admissible; their share is the endpoint-free erosion index. The certified set is unchanged at any weights-only design effect in $[0.6, 2.0]$.

country	rounds	decline $\geq$	contrasts	share
Ukraine	2–11	25.9	7/15	0.47
Cyprus	3–11	6.2	13/21	0.62
United Kingdom	1–11	6.1	11/55	0.20
Greece	1–11	3.1	8/15	0.53
Israel	1–11	2.6	10/28	0.36
Italy	1–11	1.6	5/15	0.33
Hungary	1–11	1.1	9/55	0.16
Spain	1–11	0.6	32/55	0.58

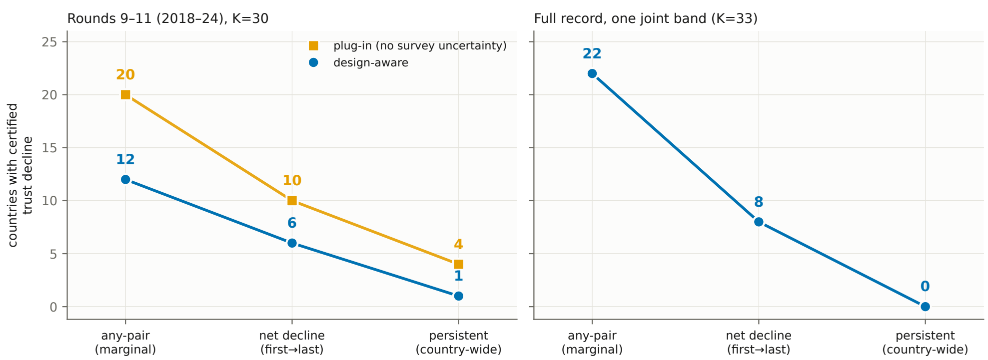

Figure 3: Certified declines collapse up the hierarchy ($\alpha = 0.10$). Left, rounds 9–11 ($K = 30$), with one path per inference layer; the headline sequence $20 \to 12 \to 6 \to 1$ reads down the plug-in marker and then along the design-aware path. Right, the full record off one joint band ($K = 33$), where the two layers coincide by construction and one path is drawn.

and net counts are unchanged when the switchers' confounded pairs are excluded, only their any-pair claims being affected. A preregistered age-group analysis likewise finds the result is not an artifact of pooling ages, youth certifying zero persistent declines as an underpowered null rather than youth stability (Supplementary Material).

A second, global target: Foa–Mounk deconsolidation. We preregistered a reanalysis of the "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the WVS/EVS trends (1981–2022, 59–77 countries with ≥ 2 waves), recoding the five-item battery so that deconsolidation is a persistent decline over each scale's preregistered core. The verdict is calibration, not denial, but the surviving phenomenon is not the one the thesis was about. A *marginal wave-pair reading of the same data*, a criterion we construct to instantiate the bottom rung, since Foa and Mounk's own empirics are cohort comparisons, flags 36–64 countries per item against 6–17 (8–24%) at the persistent rung. That 2.6–6.5$\times$ ratio compounds two changes, and we separate them: holding the inference layer fixed, the rung alone accounts for 1.7–4.8$\times$ (plug-in) or 1.9–4.8$\times$ (design-aware), and propagating design uncertainty at fixed rung accounts for the remainder. The unit is the larger effect, but not the whole of the headline number. Three caveats. The WVS ships no PSU/stratum identifiers, so the weights-only bootstrap understates design variance, which can only *inflate* counts: at variance $\times 1.5$ and $\times 2.0$ the certified core holds at 13 and 12 countries and the gap widens to 3.0–8.3$\times$, so the reported gap is a lower bound. The per-item lists carry no across-country familywise control, so we aggregate to multi-item co-certification before interpreting geography. And the certification sets sit below the $K \geq 94$ bound, so every WVS band uses the full-α anchors of Theorem 5. The youth "embrace authoritarianism" claim is only half-supported (Supplementary Material).

The certified core: where the surviving deconsolidation lives. Aggregating per-item certifications yields a positive characterization (Figure 4; `e30`): of 38 countries certified on ≥ 1 item, 13 form a certified core on ≥ 2, headed by Albania on four. The core is post-communist and Arab-Spring states with fragile Latin American and African cases alongside, not the mature democracies the thesis (Foa and Mounk, 2016, 2017) concerned, formalizing with per-country error control the decline-from-euphoric-baselines pattern the post-communist literature described qualitatively (Mishler and Rose, 1997; Inglehart and Catterberg, 2003). Three qualifications travel with it. Five core members are electoral autocracies in the V-Dem classification, where falling stated support reflects autocratic legitimation rather than deconsolidation in the consolidated-democracy sense. The West enters the core exactly twice (Finland and Switzerland, on the strongman/army pair, which we flag as a probable administration artifact), with single-item entries elsewhere: no consolidated Western democracy certifies on support for a democratic system, consistent with Wuttke et al. (2022), though not that the West is absent. And the geography is partly window-contingent: recertifying on fieldwork ending by 2017 (`e46`) keeps eight of the thirteen but drops all three Arab-Spring cases, so the post-communist component is window-stable and the Arab-Spring component rests on WVS wave 7. Certified attitudes do not forecast institutions: core membership predicts subsequent V-Dem electoral-democracy decline no better than the marginal flag ($n = 13$ versus 54; Fisher $p = 0.89$), too small a comparison to establish the decoupling recent panel work anyway expects (Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025). The estimand is the response-distribution trajectory itself: certification is descriptive error control, not a forecast. Item-level caveats are in the Supplementary Material.

8 Related work

Four literatures intersect here; the Supplementary Material gives the extended discussion. *Conformal prediction* (Vovk et al., 2005; Lei et al., 2018; Barber et al., 2023): our selector adapts not to temporal shift (Gibbs and Candès, 2021) but to the certified magnitude of survey design noise. *Cluster-as-unit conformal* (Dunn et al., 2023; Lee et al., 2023) transports guarantees to an unseen cluster but treats each cluster's data as cleanly sampled (the same clustered construction is developed, for a non-survey domain, in a companion paper (Park, 2026)); our departure is that the calibration objects are estimated under complex designs (Binder, 1983; Lumley, 2010). *Noisy calibration*: Wieczorek (2023) already brings conformal prediction to complex survey designs, for unit-level prediction under design-based inference. Our departure is the simultaneous trajectory object, the population as the exchangeable unit, and the estimated-calibration boundary. Einbinder et al. (2024); Sesia et al. (2025)

deconvolve a *known* label-noise model, and Singer et al. (2025) build conformal intervals under a known heteroskedastic measurement-error law; both rest on the identification premise that generally fails for design noise (Theorem 1), and even a supplied noise law faces the finite-K floor (Proposition 1). *Functional simultaneous bands* (Diquigiovanni et al., 2021): unlike FDR/Bonferroni corrections of marginal tests (Benjamini and Hochberg, 1995), the "persistent" object is covered directly by one band, a distinction of estimand, not error rate (Gelman et al., 2012), and the post-hoc logic by which we read a whole claim family off it is classical (Scheffé, 1953; Marcus et al., 1976; Goeman and Solari, 2011; Katsevich and Ramdas, 2020). Substantively we engage the trust-decline and deconsolidation debates (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025; Foa and Mounk, 2016, 2017; Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025; Kołczyńska and Bürkner, 2026). The closest substantive competitor is Claassen (2019, 2020), whose latent-variable panels model measurement and sampling uncertainty with model-based credible intervals whose calibration is itself contested (Tai et al., 2024). Our bands trade pooling and width for finite-sample, model-free coverage at a stated claim rung, and the two are complements. The certified core's geography intersects the post-communist disillusionment literature (Mishler and Rose, 1997; Inglehart and Catterberg, 2003), which anticipated the decline-from-euphoric-baselines pattern our certification formalizes per country.

9 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit—twice. We gave a multiplicity-honest simultaneous band for the claim unit and a hard boundary for the calibration unit: non-identified without the design-noise law (Theorem 1), unreachable at cross-national scale with it (Proposition 1). The deployed object is the reduction, a clustered band valid at any K inside a selection-free selector (Theorems 3, 5), and the design-aware layer certifies, rather than corrects, at survey scale.

When it reduces to ordinary conformal. At the national unit the method reduces exactly to clustered population conformal: the machinery certifies that the simple band is the right one here, and reports the ρ that says so. The deconvolution regime needs many exchangeable units and comparable design noise, never both across countries, but reachable one unit down (Section 6) and routine in simulation at many-unit scale (`e31`: $K \approx 220$–300 small areas, deconvolution on up to 98% of draws at 0.69–0.74× the conservative width with valid coverage).

Limitations. Four scope conditions, stated fully in the Supplementary Material. *Sampling error only:* the machinery is silent on measurement non-invariance (wording and translation changes, house effects, mode switches), which produces exactly the shifts the bands certify; certifications straddling such changes are conditional on measurement constancy. *Variance-estimator bias:* the deployed m-of-m PSU bootstrap (Rao et al., 1992; Wolter, 2007) biases design variance downward, though the Rao–Wu–Yue rescaled rerun leaves every count and gate unchanged. *A superpopulation postulate:* transport-band exchangeability is over a non-sampled, panel-attriting country set (the certifications do not use it), and holds worse for the post-communist bloc than elsewhere. *Metadata:* without design metadata (WVS) only weights-only replication is possible.

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions: persistence in none of thirty-three European democracies over two decades yet span erosion in eight, mostly invisible to adjacent-wave readings; a certified deconsolidation set several-fold smaller under a trajectory-persistence criterion than under wave-pair readings, with the surviving core concentrated outside the consolidated West. The pattern is not confined to surveys. Wherever prediction is calibrated on estimated objects—small-area model outputs, learned features, meta-analytic effects (Remark 1)—the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain when it cannot.

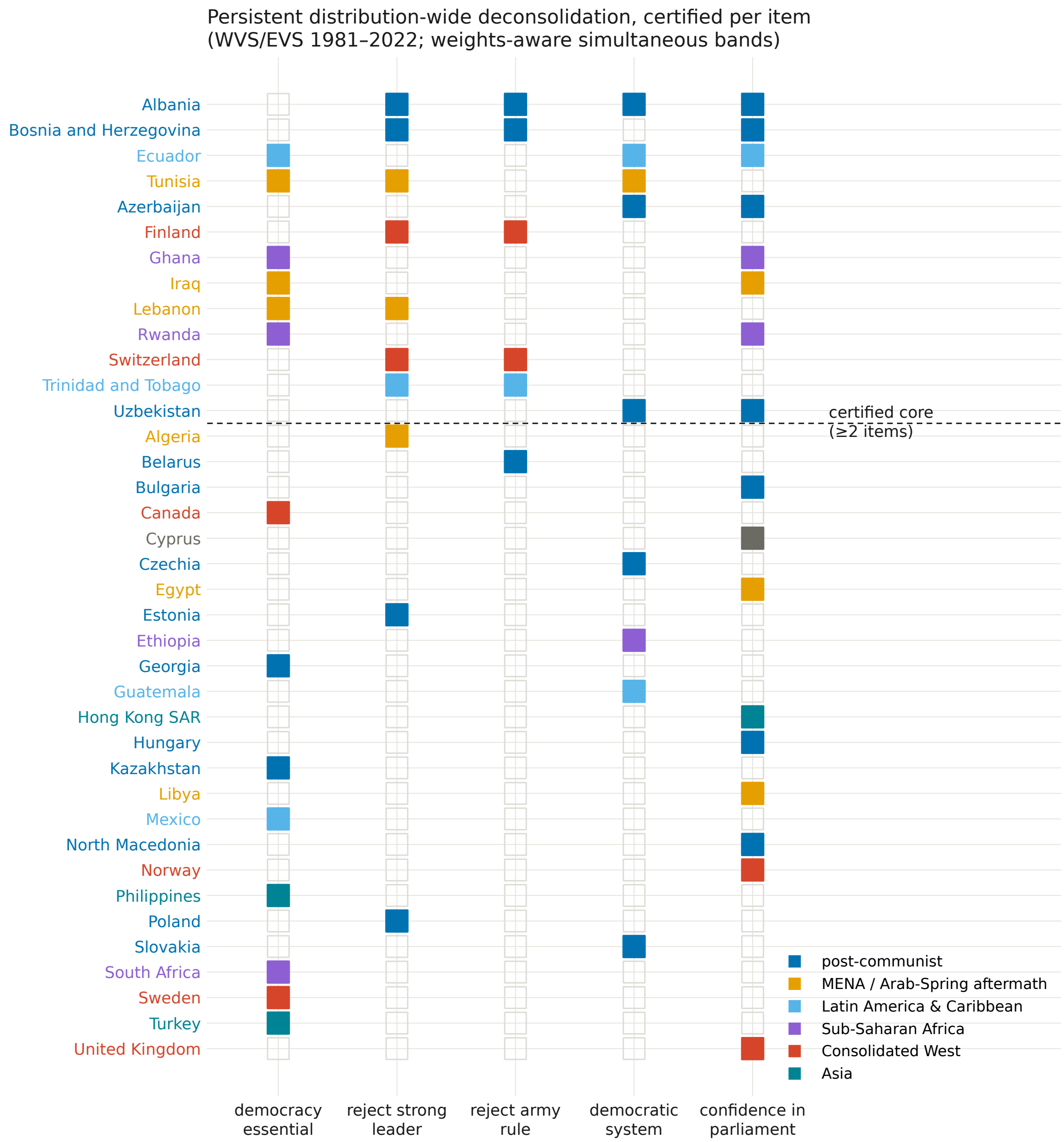

Figure 4: Co-certification across the five battery items (WVS/EVS 1981–2022; $K = 59$–77 per item). Countries above the dashed line form the ≥ 2-item certified core.

A Proofs and efficiency results

Full statements and proofs of all theorems, the efficiency results, and the modulation self-inclusion deficit are in Section S1 of the supplementary material; each theorem is paired with a machine-checked contract test.

B Frozen selector specification

The frozen constants, their development-grid derivation, and the validation protocols are in the replication package.

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Data Availability Statement

Simulation, theory, and benchmark results are deterministic under fixed seeds with no external data, and each theorem is paired with a machine-checked contract test. The ESS, World Values Survey, and AmericasBarometer/LAPOP microdata are licensed to their providers and cannot be redistributed; all code that runs on them is public, the replication package documents the exact variables, waves, and retrieval steps, and the package will be deposited in the Political Analysis Dataverse upon conditional acceptance. The editor has been notified of the restricted access at submission.

Competing Interests

The author declares none.

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